MATH3303 – Lecture 30

Prime and Irreducible Elements; $\mathbb{Z}[\sqrt{-5}]$

Review Notes

Reference: Lee, Abstract Algebra*, Section 10.3*

Overview

This lecture introduces two parallel notions – *prime* and *irreducible* elements – and shows they generally differ. It then sets up the structural lemma that will close the gap in a PID.

- **Prime elements** and the correspondence: p prime $\iff$ (p) a prime ideal
- **Irreducible elements** and the theorem: every prime is irreducible
- $\mathbb{Z}[\sqrt{-5}]$**:** the standard counter-example where 3 is irreducible but not prime
- **Lemma (in a PID):** r irreducible $\Rightarrow$ (r) maximal

Connection to Lecture 29. Lecture 29 introduced PIDs and the ACC. This lecture takes the next step: we need a workable notion of "building block" for factorisation. There are two candidates – prime and irreducible elements – which coincide in $\mathbb{Z}$ but generally do not (witness $\mathbb{Z}[\sqrt{-5}]$). Lecture 31 will use the lemma below to prove they coincide in any PID.

1 Prime Elements

Definition 1.1: Prime element (from lecture; cf. Lee, Def. 10.10)

Let R be an integral domain and $p \in R \setminus \{0\}$. We say p is *prime* if p is not a unit, and if $p \mid ab$ then $p \mid a$ or $p \mid b$.

Lemma 1.1: Prime element $\iff$ prime ideal (from lecture)

Let R be an integral domain and $p \in R \setminus \{0\}$. Then

$$p \text{ is prime} \iff (p) \text{ is a prime ideal.}$$

Proof. *($\Rightarrow$)* Let p be prime. Then $(p) \neq R$, since otherwise $1 \in (p)$ would force p to be a unit – contradiction.

Assume $xy \in (p)$. Then $p \mid xy$, so $p \mid x$ or $p \mid y$, i.e. $x \in (p)$ or $y \in (p)$. Hence (p) is prime.

($\Leftarrow$) Let (p) be a prime ideal. Then p is not a unit (otherwise $(p) = R$). If $p \mid xy$, then $xy \in (p)$, so $x \in (p)$ or $y \in (p)$, i.e. $p \mid x$ or $p \mid y$. $\square$

2 Irreducible Elements

Definition 2.1: Irreducible element (from lecture; cf. Lee, Def. 10.11)

Let R be an integral domain and $p \in R \setminus \{0\}$. We say p is *irreducible* if p is not a unit, and if $p = ab$ then a or b is a unit.

Theorem 2.1: Every prime is irreducible (from lecture; cf. Lee, Thm. 10.10)

Let R be an integral domain. Then every prime element of R is irreducible.

Proof. Let p be prime and $a, b \in R$ with $p = ab$.

Aim: either a or b is a unit.

$p \mid ab$, so $p \mid a$ or $p \mid b$. WLOG $p \mid a$, so $a = px$ for some $x \in R$. Then

$$p = ab = pxb \implies p(1 - xb) = 0.$$

Since $p \neq 0$ and R is an integral domain, $1 - xb = 0$, i.e. $xb = 1$. So b is a unit. Hence p is irreducible. $\square$

Remark 2.1: The converse is NOT true in general (from lecture)

There exist integral domains containing irreducible elements that are not prime. The canonical example is $\mathbb{Z}[\sqrt{-5}]$, treated in the next section.

3 The Norm on $\mathbb{Z}[\sqrt{-5}]$

Definition 3.1: (from lecture)

Consider the integral domain

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + \sqrt{-5}\, b : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}.$$

Just like for $\mathbb{Z}[i]$, define a norm $N : \mathbb{Z}[\sqrt{-5}] \to \mathbb{Z}_{\geq 0}$ by

$$N(\alpha) := \alpha\, \overline{\alpha}.$$

For $\alpha = a + \sqrt{-5}\, b$,

$$N(\alpha) = (a + b\sqrt{-5})(a - b\sqrt{-5}) = a^2 + 5b^2.$$

N is multiplicative:

$$N(\alpha\beta) = \alpha\beta\, \overline{\alpha\beta} = \alpha\overline{\alpha} \cdot \beta\overline{\beta} = N(\alpha)N(\beta).$$

Example 3.1: 3 is irreducible in $\mathbb{Z}[\sqrt{-5}]$ but not prime (from lecture; cf. Lee, Example 10.15)

(1) 3 is not prime. We have

$$3 \mid 9 = (2 + \sqrt{-5})(2 - \sqrt{-5}),$$

since $9 = 4 - 5 \cdot (-1)$. But $3 \nmid (2 + \sqrt{-5})$: if $2 + \sqrt{-5} = 3(a + b\sqrt{-5})$ for some $a, b \in \mathbb{Z}$, then $2 = 3a$ and $1 = 3b$, impossible. Similarly $3 \nmid (2 - \sqrt{-5})$.

(2) $\alpha \in R$ is a unit $\iff N(\alpha) = 1 \iff \alpha = \pm 1$.
($\Rightarrow$) If α is a unit, $\exists\, \beta$ with $\alpha\beta = 1$. Then $N(\alpha)N(\beta) = N(1) = 1$, so $N(\alpha) = N(\beta) = 1$.
($\Leftarrow$) $N(\alpha) = a^2 + 5b^2 = 1$ forces $b = 0$ and $a = \pm 1$. Both ± 1 are units.

(3) 3 is irreducible. Suppose $3 = \alpha\beta$ for $\alpha, \beta \in R$. Then

$$9 = N(3) = N(\alpha)N(\beta).$$

Case analysis on $N(\alpha)N(\beta) = 9$:

- $N(\alpha) = N(\beta) = 3$: $a^2 + 5b^2 = 3$ has no integer solutions. **Impossible.**
- $N(\alpha) = 1, N(\beta) = 9$ (or vice versa): then α (resp. β) is a unit by (2).

Hence 3 is irreducible.

So 3 is irreducible but not prime, witnessing the failure of the converse to Theorem 2.2 in general.

Remark 3.1: Consequence (from lecture)

$\mathbb{Z}[\sqrt{-5}]$ is not a PID. (We prove the contrapositive in the next section: in a PID, every irreducible is prime.)

4 In a PID: Irreducible $\Rightarrow$ (r) Maximal

Lemma 4.1: Irreducible $\iff$ (r) maximal in a PID – $(\Rightarrow)$ direction (from lecture)

Let R be a PID and $r \in R$. Then

$$r \text{ is irreducible} \implies (r) \text{ is a maximal ideal.}$$

Proof. Let I be an ideal of R with $(r) \subseteq I \subseteq R$. Since R is a PID, $I = (a)$ for some $a \in R$. Since $(r) \subseteq (a)$, $r \in (a)$, so $r = ab$ for some $b \in R$. Since r is irreducible, either a or b is a unit.

- If a is a unit, then $I = (a) = R$.
- If b is a unit, then $a = rb^{-1} \in (r)$, so $I = (a) \subseteq (r)$. Combined with $(r) \subseteq I$, $I = (r)$.

Therefore (r) is maximal. $\square$

Remark 4.1: Looking ahead (from lecture)

The reverse implication (r) maximal $\Rightarrow$ r irreducible is proved at the start of Lecture 31. The combined lemma is then used to show that in a PID, irreducible $\iff$ prime, closing the gap exposed by $\mathbb{Z}[\sqrt{-5}]$.

Checklist of Skills

- ☐ State the definition of a prime element in an integral domain.
- ☐ Prove that p is prime $\iff$ (p) is a prime ideal.
- ☐ State the definition of an irreducible element.
- ☐ Prove that every prime element of an integral domain is irreducible.
- ☐ Define and use the multiplicative norm $N(\alpha) = a^2 + 5b^2$ on $\mathbb{Z}[\sqrt{-5}]$.
- ☐ Show that $\mathbb{Z}[\sqrt{-5}]^\times = \{\pm 1\}$.
- ☐ Prove that 3 is irreducible but not prime in $\mathbb{Z}[\sqrt{-5}]$.
- ☐ Prove that in a PID, every irreducible element generates a maximal ideal.